

SUMMARIZER

Academia de Código



Fausto Nunes - 07/02/2023



Exceptions

Let's make our code more
professional!

Simple Gfx

Let's make something fun!

EXCEPTIONS

We have to catch bugs... and exceptions!



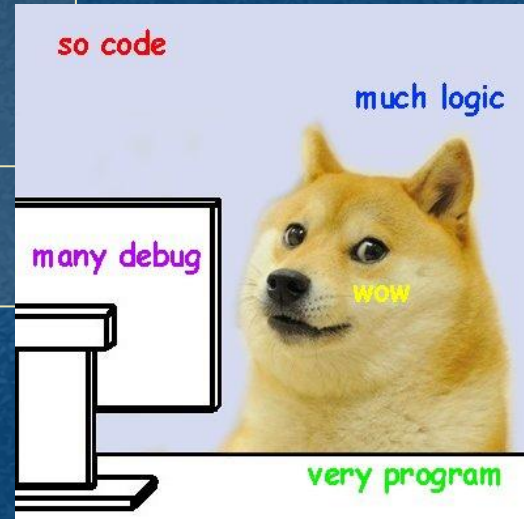
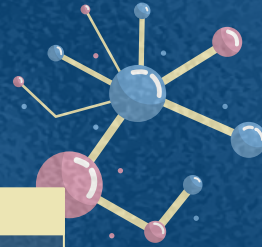


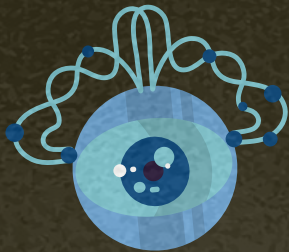
WHΘA

The next slide has the obligatory boring definition of
Exceptions

“**Exception** is an unwanted or unexpected event, which occurs during the execution of a program, that disrupts the normal flow of the program’s instructions.”

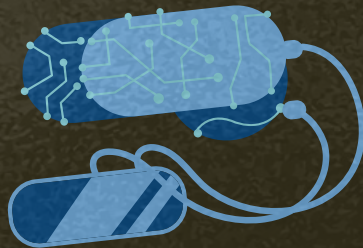
— Our dear friend Google





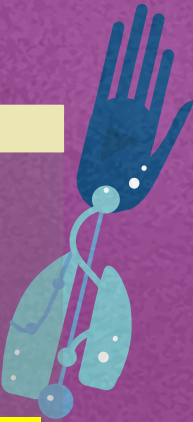
Long story short...

Exceptions BREAK OUR PROGRAM(S) so we should provide a better way of handling them to keep things going smoothly.





Obligatory
joke incoming





**try
and
catch**

exceptions

Reasons why exceptions occur:

Invalid user input

Device failure

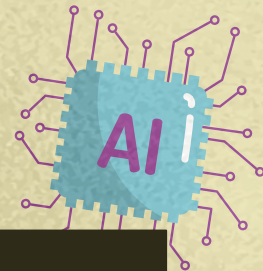
Loss of network
connection

Physical
limitations

Code errors

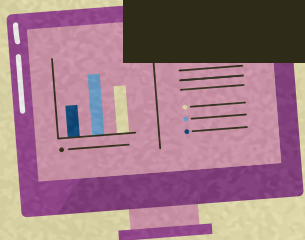
Opening an
unavailable file

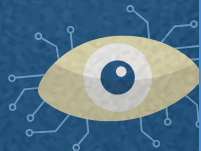
Etc.



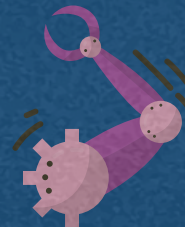
An example

A quick and easy one.



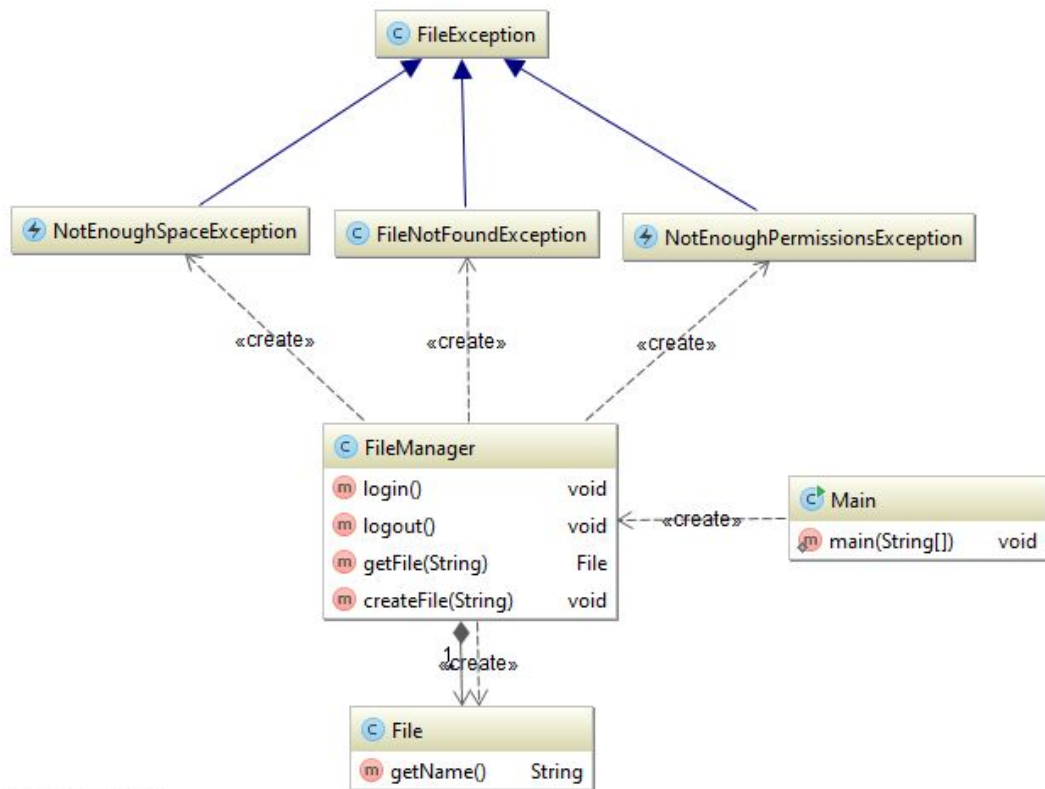


```
try {  
  
    System.out.println("Enter a whole number to divide: ");  
    int x = 5;  
  
    System.out.println("Enter a whole number to divide by: ");  
    int y = 0;  
  
    int z = x/y;  
  
    System.out.println("result: " + z);  
}  
  
catch(ArithmeticException e) {  
    System.out.println("You can't divide by zero! IDIOT!");  
}
```





Exercise





```
public class FileException extends Exception {  
    private String exceptionName;  
  
    public FileException(String name){  
        this.exceptionName = name;  
    }  
    public String getExceptionName() {  
        return exceptionName;  
    }  
}
```



```
public class FileManager {  
  
    public void createFile(String name) throws FileNotFoundException {  
        if(!isLoggedIn) throw new NotEnoughPermissionsException();  
        for(int i = 0; i < fileStorage.length; i++){  
            if(fileStorage[i] == null){  
                fileStorage[i] = new File(name);  
                return;  
            }  
        }  
  
        throw new NotEnoughSpaceException();  
    }  
}
```

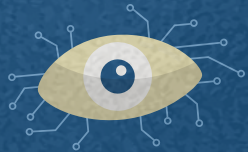


```
public class NotEnoughSpaceException extends FileException{  
  
    public NotEnoughSpaceException(){  
        super("You don't have enough space.");  
    }  
  
}
```

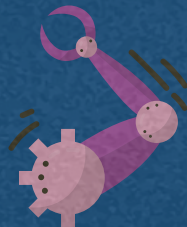


```
public class Main {  
    public static void main(String[] args) {  
  
        FileManager fileManager = new FileManager();  
  
        try {  
            fileManager.login();  
            //fileManager.login();  
            fileManager.createFile("lol.jpeg");  
            fileManager.createFile("spotify");  
            //fileManager.createFile("error");  
            fileManager.getFile("spotify");  
            //fileManager.getFile("virus");  
            fileManager.logout();  
            //fileManager.logout();  
            fileManager.createFile("abc");  
            fileManager.getFile("spotify");  
  
        } catch (FileNotFoundException e) {  
            System.out.println(e.getExceptionName());  
        }  
    }  
}
```

Access Level Modifiers



Try to keep things as private as possible! Cof cof 🤪



Simple GFX

Yay pictures moving and stuff!

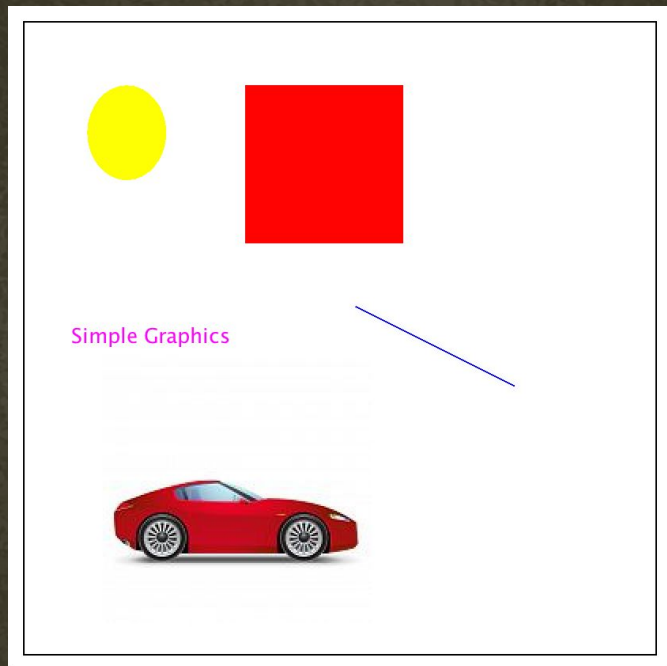


Simple GFX

Graphical library created
by Academia de Código
purely for our enjoyment.



Simple GFX

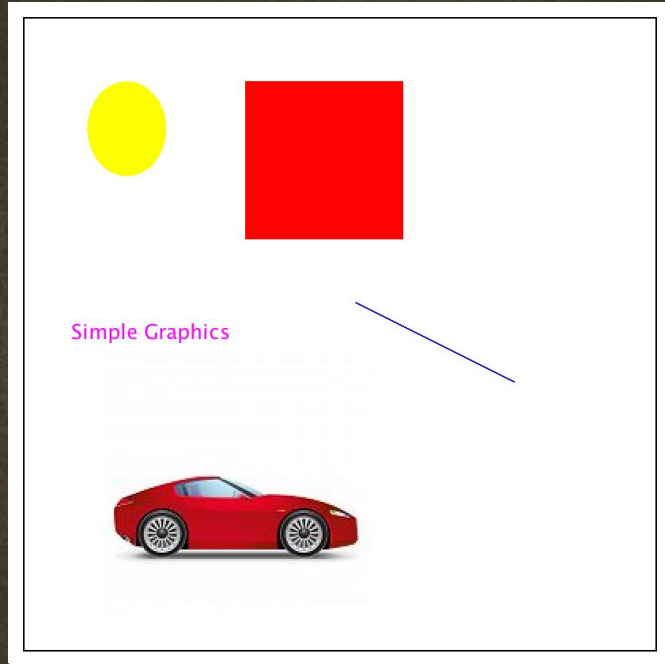


It creates:

- Rectangles
- Ellipses
- Text
- Lines

● Images

Simple GFX



Takes input from:

- Keyboard
- Mouse

Simple GFX

```
public class Main {  
    public static void main(String[] args) {  
  
        System.out.println("Much graphics. Such wow.");  
  
        MagicKeyboard magicKeyboard = new MagicKeyboard();  
        magicKeyboard.init();  
  
        Picture fanuca = new Picture(600,598,"resources/fanuca.png");  
        fanuca.draw();  
  
        Picture oldGuy = new Picture(900,526, "resources/oldguy.png");  
        oldGuy.draw();  
  
    }  
}
```

Simple GFX

```
public class Player {  
  
    private Picture picture;  
  
    public Player(String filepath) {  
        picture = new Picture(200,598, filepath);  
        picture.draw();  
    }  
  
    public void moveRight(){  
        picture.translate(10,0);  
    }  
  
    public void moveLeft(){  
        picture.translate(-10,0);  
    }  
  
    public void moveUp(){  
        picture.translate(0,-10);  
    }  
  
    public void moveDown(){  
        picture.translate(0,10);  
    }  
}
```

Simple GFX



That's it...

THANKS!

